

Thomas Debris-Alazard

BORN IN PARIS, FRANCE, MAY 1, 1991 · RESEARCHER SCIENTIST AT INRIA

☎ (+33) 631053595 | ✉ thomas.debris@inria.fr | 🌐 <http://tdalazard.io/>

Research Interests

Public-Key Cryptography with a focus on code- and lattice-based cryptography

- **Cryptographic designs, cryptanalysis, reductions**
- **Algorithms** classical and quantum
- **Security estimates** study of generic decoding problems
- **Security proofs** in the classical or quantum model

Employment

École Polytechnique

TEACHER ASSISTANT (CHARGÉ D'ENSEIGNEMENT)

Département d'Informatique de l'École Polytechnique (DIX)

Saclay, France

Sept. 2022 - Present

Inria Saclay

RESEARCHER SCIENTIST (CHARGÉ DE RECHERCHE)

Project-Team: Grace

Saclay, France

Sept. 2020 - Present

Royal Holloway, University of London, UK

POSTDOC IN THE INFORMATION SECURITY GROUP

Hosted by Pr Martin R. Albrecht

London, UK

Sept. 2019 - Sept. 2020

Education

Inria Paris

PH.D., CODE-BASED CRYPTOGRAPHY: NEW APPROACHES FOR DESIGN AND PROOF ; CONTRIBUTION TO

CRYPTANALYSIS

Advisor: Pr Jean-Pierre Tillich

Paris, France

Sept. 2016 - Sept. 2019

École Normale Supérieure de Cachan (ENS)

THESIS, CODE-BASED CRYPTOGRAPHY: STUDY OF A GENERIC DECODING ALGORITHM, STATISTICAL DECODING

Advisor: Pr Jean-Pierre Tillich

MASTER MPRI (PARISIAN MASTER OF RESEARCH IN COMPUTER SCIENCE).

Main Topics: Cryptography, Complexity, Security reductions, Gröebner basis, Quantum algorithms

AGRÉGATION DE MATHÉMATIQUES OPTION INFORMATIQUE.

Paris, France

Mar. 2016 - Sept. 2016

Sept. 2015 - Sept. 2016

Sept. 2014 - Sept. 2015

Honors and Awards

2026-2031 **ERC Starting Grant**

IQ-SCALE: IRONCLAD QUANTUM SECURITY OF CODE- AND LATTICE-BASED CRYPTOGRAPHY

1.5M€

2021-2025 **ANR JCJ**

COLA: AN INTERFACE BETWEEN CODE AND LATTICE-BASED CRYPTOGRAPHY

200 k€

2021 **Finalist for the Cor Baayen Young Researcher Award**

ERCIM

2020 **Gilles Kahn Thesis Award**

THOMAS DEBRIS-ALAZARD UNDER THE SUPERVISION OF JEAN-PIERRE TILlich

Société Informatique de France

2019 **Best Paper Award, Wave: A New Family of Trapdoor One-Way Preimage Sampleable Functions Based on Codes**

THOMAS DEBRIS-ALAZARD, NICOLAS SENDRIER AND JEAN-PIERRE TILlich

Asiacrypt '19

Scientific Publications

- 2025 **MIRANDA: short signatures from a leakage-free full-domain-hash scheme** *Asiacrypt '26*
ALAIN COUVREUR, THOMAS DEBRIS-ALAZARD, PHILIPPE GABORIT, AND ADRIEN VINCOTTE
- 2025 **Hardness of Problems with Hints in Code-Based Cryptography and Applications to Traitor Tracing** *Asiacrypt '26*
THOMAS DEBRIS-ALAZARD, VICTOR DYSERYN AND DUONG HIEU PHAN
- 2026 **A Minrank-based encryption scheme à la Alekhnovich-Regev** *Eurocrypt '26*
THOMAS DEBRIS-ALAZARD, PHILIPPE GABORIT, ROMARIC NEVEU AND OLIVIER RUATTA
- 2025 **Worst and Average Case Hardness of Decoding via Smoothing Bounds** *PKC '25*
THOMAS DEBRIS-ALAZARD AND NICOLAS RESCH
- 2024 **New Solutions to Delsarte's Dual Linear Programs** *IEEE Information Theory '24*
ANDRÉ CHAILLOUX AND THOMAS DEBRIS-ALAZARD
- 2024 **Exploiting signature leakages: breaking Enhanced pqsigRM** *ISIT '24*
THOMAS DEBRIS-ALAZARD, PIERRE LOISEL AND VALENTIN VASSEUR
- 2024 **Quantum Oblivious LWE Sampling and Insecurity of Standard Model Lattice-Based SNARKs** *STOC '24*
THOMAS DEBRIS-ALAZARD, POURIA FALLAHPOUR AND DAMIEN STEHLÉ
- 2024 **Reduction from sparse LPN to LPN, Dual Attack 3.0** *Eurocrypt '24*
KEVIN CARRIER, THOMAS DEBRIS-ALAZARD, CHARLES MEYER-HILFIGER AND JEAN-PIERRE TILlich
- 2023 **Quantum Reduction of Finding Short Code Vectors to the Decoding Problem** *IEEE Information Theory '23*
THOMAS DEBRIS-ALAZARD, MAXIME REMAUX AND JEAN-PIERRE TILlich
- 2023 **On the pseudorandomness of the decoding problem via the Oracle Comparison Problem** *Asiacrypt '23*
MAXIME BOMBAR, ALAIN COUVREUR AND THOMAS DEBRIS-ALAZARD
- 2023 **Smoothing codes and lattices: systematic study and new bounds** *IEEE Information Theory '23*
THOMAS DEBRIS-ALAZARD, LÉO DUCAS, NICOLAS RESCH AND JEAN-PIERRE TILlich
- 2022 **Statistical Decoding 2.0: Reducing Decoding to LPN** *Asiacrypt '22*
KEVIN CARRIER, THOMAS DEBRIS-ALAZARD, CHARLES MEYER-HILFIGER AND JEAN-PIERRE TILlich
- 2022 **On Codes and Learning with Errors over Function Fields** *Crypto '22*
MAXIME BOMBAR, ALAIN COUVREUR AND THOMAS DEBRIS-ALAZARD
- 2022 **An Algorithmic Reduction Theory for Binary Codes: LLL and more** *IEEE Information Theory '22*
THOMAS DEBRIS-ALAZARD, LÉO DUCAS AND WESSEL P.J. VAN WOERDEN
- 2021 **Classical and Quantum algorithms for generic Syndrome Decoding problems and applications to the Lee metric** *PQCrypto '21*
ANDRÉ CHAILLOUX, THOMAS DEBRIS-ALAZARD AND SIMONA ETINSKI
- 2020 **Tight and Optimal Reductions for Signatures based on Average Trapdoor Preimage Sampleable Functions and Applications to Code-Based Signatures** *PKC '20*
ANDRÉ CHAILLOUX AND THOMAS DEBRIS-ALAZARD
- 2019 **Wave: A New Family of Trapdoor One-Way Preimage Sampleable Functions Based on Codes** *Asiacrypt '19*
THOMAS DEBRIS-ALAZARD, NICOLAS SENDRIER AND JEAN-PIERRE TILlich
- 2019 **Ternary syndrome decoding with large weights** *SAC '19*
RÉMI BRICOUT, ANDRÉ CHAILLOUX, THOMAS DEBRIS-ALAZARD AND MATTHIEU LEQUESNE

2018	Two attacks on rank metric code-based schemes: Ranksign and an identity-based-encryption scheme THOMAS DEBRIS-ALAZARD AND JEAN-PIERRE TILLICH	<i>Asiacrypt '18</i>
2017	Statistical Decoding THOMAS DEBRIS-ALAZARD AND JEAN-PIERRE TILLICH	<i>ISIT '17</i>

Preprints

2021	Wavelet: Code-based postquantum signatures with fast verification on microcontrollers GUSTAVO BANEGAS, THOMAS DEBRIS-ALAZARD, MILENA NEDELJKOVIĆ AND BENJAMIN SMITH	<i>iacr.org</i>
2020	On the Hardness of Code Equivalence Problems in Rank Metric ALAIN COUVREUR, THOMAS DEBRIS-ALAZARD AND PHILIPPE GABORIT	<i>arxiv.org</i>
2019	About Wave Implementation and its Leakage Immunity THOMAS DEBRIS-ALAZARD, NICOLAS SENDRIER AND JEAN-PIERRE TILLICH	<i>iacr.org</i>
2017	The problem with the SURF scheme THOMAS DEBRIS-ALAZARD, NICOLAS SENDRIER AND JEAN-PIERRE TILLICH	<i>arxiv.org</i>

Teaching

PhD. Supervision

2023-	Pierre Loisel ON CODE ALGORITHMS AND CRYPTANALYSIS	<i>with Alain Couvreur</i>
2020-2023	Maxime Bombar ON STRUCTURES CODES IN CRYPTOGRAPHY (DEFENDED ON DECEMBER 15, 2023)	<i>with Alain Couvreur</i>

Courses

2024-	Advanced topics in quantum information and computing (CSC_51002_EP) ÉCOLE POLYTECHNIQUE
2023-	Introduction to information theory (INF563, CSC_51063_EP) ÉCOLE POLYTECHNIQUE
2022-	Introduction to quantum computing and quantum information (INF587, MDC_51002_EP) ÉCOLE POLYTECHNIQUE
2021-	Error-correcting codes and applications to cryptography MPRI, WITH ANNE CANTEAUT AND ALAIN COUVREUR
2021-2023	Post-quantum cryptography, introduction to code-based cryptography ENS LYON, WITH DAMIEN STEHLÉ AND BENJAMIN WESOŁOWSKI

Tutorials

Sept. 2024	Summer School IES Corsica, INTRODUCTION TO CODE-BASED CRYPTOGRAPHY	<i>Cargèse</i>
June. 2024	Introduction to Quantum-Safe Cryptography (IBM Zurich) INTRODUCTION TO CODE-BASED CRYPTOGRAPHY	<i>Zurich</i>
Oct. 2023	CIMPA school: mathematical aspects of post-quantum cryptography, INTRODUCTION TO CODE-BASED CRYPTOGRAPHY	<i>Rabat</i>
Aug. 2022	Summer school in post-quantum cryptography, INTRODUCTION TO CODE-BASED CRYPTOGRAPHY	<i>Budapest</i>

Invited Talks

2024 **Mathematics for post-quantum cryptanalysis** *Budapest*

2024 **Thirteenth in the series workshop Coding and Cryptography (WCC)** *Perugia*

Program Committees

- 2021- **Program committee member**
GILLES THESIS KAHN AWARD '21-23, EUROCRYPT '25, PKC '25, SAC '25, CRYPTO '25, PQCRYPTO '26
- 2025- **Editorial board member**
DESIGNS, CODES AND CRYPTOGRAPHY (DCC)

Presentations

Selected Talks at Seminars, Workshops and Conferences

- Dec, 2024 **Codes and Lattices in Cryptography: real twins or distant cousins?** CAIPI WORKSHOP *Limoges*
- Feb, 2024 **Codes and Lattices in Cryptography: real twins or distant cousins?** ATTACC WORKSHOP *Munich*
- Sept, 2023 **Wave: a Code-based Hash and Sign Signature Scheme,** OXFORD POST-QUANTUM CRYPTOGRAPHY SUMMIT (PQCS) *Oxford*
- Oct, 2021 **Quantum Reduction of Finding Short Code Vectors to the Decoding Problem,** DAGSTUHL SEMINAR, QUANTUM CRYPTANALYSIS *Dagstuhl*
- Dec, 2019 **Wave: A New Family of Trapdoor One-Way Preimage Sampleable Functions Based on Codes,** ASIACRYPT 19' *Kobe*
- Sept, 2019 **Wave: A New Family of Trapdoor One-Way Preimage Sampleable Functions Based on Codes,** LONDON-ISH LATTICE CODING AND CRYPTO MEETINGS *Imperial College, London*
- May, 2019 **Wave: A New Family of Trapdoor One-Way Preimage Sampleable Functions Based on Codes,** CRYPTO MEETING *ENS, Lyon*
- Feb, 2019 **Wave: A New Family of Trapdoor One-Way Preimage Sampleable Functions Based on Codes,** CRYPTOGRAPHY SEMINAR *PQShield, Oxford*
- Dec, 2018 **Two attacks on rank metric code-based schemes: Ranksign and an identity-based-encryption scheme,** ASIACRYPT 18' *Brisbane*
- June, 2017 **Statistical Decoding,** ISIT 17' *Aachen*

Workshops

- 2020-2024 **Organization of the team Grace Seminar,** PRESENTATIONS: HERE *Inria Saclay*
- 2020-2022 **Workshop on Transference,** ORGANIZED BY LÉO DUCAS PRESENTATION: SMOOTHING BOUNDS FOR CODES AND LATTICES *CWI*

2019-2020 **Workshop “yet another crypto reading group”**, ORGANIZED BY MARTIN R. ALBRECHT

*Royal Holloway
University of London*

PRESENTATION: WORST-CASE HARDNESS FOR LPN AND CRYPTOGRAPHIC HASHING VIA CODE SMOOTHING

2016 - **Workshop “code-based cryptography”**, ORGANIZED BY JEAN-PIERRE TILICH

Inria Paris

PRESENTATIONS: QUANTUM OBLIVIOUS LWE SAMPLING, ON THE PSEUDORANDOMNESS OF THE DECODING PROBLEM VIA THE ORACLE COMPARISON PROBLEM, STATISTICAL DECODING, SURF : A NEW CODE-BASED SIGNATURE SCHEME, TWO ATTACKS AGAINST SCHEMES BASED ON RANK METRIC, NEW RESULTS ABOUT SIGNATURES BASED ON CODES, WAVE, WORST-CASE HARDNESS FOR LPN AND CRYPTOGRAPHIC HASHING VIA CODE SMOOTHING, AN ALGORITHMIC REDUCTION THEORY FOR BINARY CODES: LLL AND MORE, QUANTUM REDUCTION OF FINDING SHORT CODE VECTORS TO THE DECODING PROBLEM, SMOOTHING BOUNDS: FROM LATTICES TO CODES AND BACK TO LATTICES

Scientific Popularization

2021	Rendez-vous des Jeunes Mathématiciennes et Informaticiennes, Fête de la science à l'école Polytechnique, Olympiades de Mathématiques de l'Académie de Créteil
2018	International Tournament of Young Mathematicians (Jury Member)
2018	Tournoi Français des Jeunes Mathématiciennes et Mathématiciens (Jury Member)
2018	Rendez-vous des Jeunes Mathématiciennes et Informaticiennes